

Lesson 12: The National Grid and exam clinic

Electric Circuits · Year 10 · 50 minutes · AQA 6.2.4.3; 6.2.1-4

Why this lesson is here

A power station sends energy over long cables. A step-up transformer raises p.d. so the same power can be transmitted with a smaller current.

Learning objective

Explain efficient transmission and tackle mixed AQA-style electricity questions.

Core teaching point

For the same transmitted power, raising V lowers I ($P = VI$); smaller I greatly reduces cable heating ($P = I^2R$). Step down before domestic use.

Equipment and preparation

Grid flow diagram, calculators, equation toolkit, end-of-unit exam clinic.

Lesson sequence

Time	Phase	Teacher move and evidence
0-6	Retrieval	Explain $P = VI$ and I^2R in one sentence each.
6-18	Model	Grid sequence and step-up reasoning.
18-28	Calculate	Work through current and loss ratios; HT transformer task.
28-44	Exam clinic	Attempt selected questions 1, 3, 7 and 8; finish the rest at home and self-correct with guidance.
44-50	Exit	Set one content target and one exam technique target.

Workbook task guidance

Task 1: Draw a flow from power station to step-up transformer, transmission cables, step-down transformer and home; label why each step matters.

Expected: Power station → step-up → high-p.d. cables → step-down → homes; note lower transmission current.

Task 2: Use $P = VI$ to compare currents for 1.0 MW transmitted at 10 kV and 100 kV; then compare I^2R loss for the same cable R .

Expected: $I = 100$ A at 10 kV and 10 A at 100 kV; with same R , cable loss falls by factor 100.

Task 3: HT: calculate secondary current for an ideal transformer with 20 kV, 500 A input and 200 kV output.

Expected: HT: $I_s = (20000 \times 500)/200000 = 50$ A.

Task 4: Start the end-of-unit exam clinic in class. Complete remaining questions at home, then correct in a different colour using the mark guidance.

Expected: Use the exam clinic marking notes after completing all questions; record one content and one technique target.

Worked example

HT: ideal transformer has 25 kV, 400 A in and 400 kV out. $I_s = V_p I_p / V_s = 25$ A.

AQA-style checkpoint

[4 marks] Explain why the National Grid transmits electricity at high p.d. and why the p.d. is reduced before homes.

Mark guidance: Step-up raises p.d. (1); for fixed power, current falls (1); smaller I means much less I^2R heating/loss in cables (1); step-down gives lower p.d. suitable for consumers (1).

Anticipated misconception

A step-up transformer changes p.d. and current; it does not create extra power.

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.

Exam clinic mark guide

1: $2.5 \text{ min} = 150 \text{ s}$; $Q = It = 0.80 \times 150 = 120 \text{ C}$ (3).

2: $V = E/Q = 48/8.0 = 6.0 \text{ V}$; 6 J transferred per coulomb (4).

3: $R_{\text{total}} = 12 \text{ ohm}$; $I = 1.0 \text{ A}$; each p.d. = 6.0 V (4).

4: Contact position/reading error, heating or poor connection; repeat point, check apparatus, compare with trend (any well-justified four points).

5: $R = 10 \text{ ohm}$ then 20 ohm ; current heats filament and its resistance increases (4).

6: Earth connects metal case to 0 V via low-resistance path; fault gives large current; fuse melts/breaker trips and disconnects live, protecting user (5).

7: $P = 1500 \text{ W}$, $t = 240 \text{ s}$, $E = 360000 \text{ J}$; $I = P/V = 1500/230 = 6.5 \text{ A}$ (5).

8: High V lowers I for fixed P ($P=VI$); lower I reduces I^2R heating; step down for users. HT: $I_s = (20000 \times 500)/200000 = 50 \text{ A}$ (7).

Physics-only static electricity extension

AQA separate Physics 4.2.5 is outside Trilogy and the site's eight-mission scope. Workbook answers: electrons moved cloth to rod, leaving cloth positive; field arrows point radially out from a positive sphere, denser near it; the negative sphere is attracted by the positive sphere's field, and the force increases as separation decreases. Sparking is charge moving through air.